Characterize Before You Loop: a Zero-Shot Proposer's Template Does Not Survive the Loop's Own Conditions

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Abstract

A weak-tier language model proposer asked in a zero-shot call to pack circles in the unit square concentrates on a grid-plus-filler template. A companion submission establishes that concentration and gives the closed form that names the modal output in advance. This paper measures what happens to the characterization when the setting changes. Eight registered arms change four things about it: the parent, the iteration, the serving path with its reasoning setting, and the vendor. One parent-conditioning step dissolves the template on both halves of the registered rule. Five generations of a minimal archive-and-mutate loop, at population 5 and 49 valid outputs from 100 conditioned invocations, do not restore it at that scale, and two of the loop's four predictions were not met. The same prompts on a pinned serving path return no valid packing at the square cells, and a registered follow-up names extended thinking, on against off at the routing layer's default effort, as the component that restores validity, at one cell on 14 and 13 rows. At a second vendor the pooled prediction holds. At the cells that can discriminate anchoring from family search it does not. Across thirteen free-tier aliases, ten fall below arm V's registered floor of 10 valid outputs in 25 invocations, two transfer at point estimate, at 8 of 12 and 11 of 18 with Wilson intervals that include 50%, and one does not transfer; all three evaluable aliases were added by a registered amendment. The template a weak-tier LLM proposer concentrates on in zero-shot circle packing is contingent on the serving path and the request parameters it carries, the reasoning setting included; it is neither confined to one instrument nor invariant across them. A zero-shot characterization of a proposer therefore licenses no prediction about the same proposer inside a discovery loop without a measurement at that loop's own conditions. Every registered arm is reported, whichever way each went.

1 Introduction

Ask a language model to place 21 circles in a unit square to maximize the sum of their radii, and it will not search. It reaches for the nearest square grid, five by five ($k = 5$) at radius $1/(2k)$, and truncates it to 21 circles, leaving four cells empty. That behaviour is sharp enough to write down. A companion submission (Anonymous authors, 2026) reports the closed form that names the modal output in advance. It reports that form as an empirical mode at all seven tested N and at four of five held-out N. It also finds the form under a perturbed container and under paraphrase. This paper's own generation-0 rows, byte-identical to those unconditioned calls (Anonymous authors, 2026), put the predicted value on valid outputs at $N = 13$ and $N = 31$ at the rates the arm L row of Table 1 reports.

A characterization that sharp invites an inference. If the proposer emits a computable value in a zero-shot call, a study can treat that value as what the proposer will contribute inside a discovery loop. It can then reason about the loop without running it. LLM-driven discovery systems in the FunSearch line (Romera-Paredes et al., 2024; Novikov et al., 2025) do not call their proposer zero-shot. They condition it on a parent and iterate it against an archive. Each of those is a change of setting, and each is a place the characterization can fail. So is the serving path, which the companion paper reports as an instrument the characterization is scoped to (Anonymous authors, 2026).

This paper measures the failures. Eight registered arms are reported here, and between them they change four things about the setting. Arm MU changes the parent the proposer is conditioned on. Arm L changes the iteration, feeding a scored proposal back as the next parent. Arms P and P-D change the serving path and the reasoning setting on it. Arms GM, GM2, GM3 and V change the vendor, and GM2 and GM3 differ only by output budget at that vendor, 4,096 tokens against 16,384. GM and GM2 returned no scoreable data, and are reported because they were registered and because GM2's failure diagnosed the output budget that arm GM3 corrects. Every registered prediction, verdict and falsifier is reported whichever way it went.

Claim. The template a weak-tier LLM proposer concentrates on in zero-shot circle packing is contingent on the serving path and the request parameters it carries, the reasoning setting included; it is neither confined to one instrument nor invariant across them. Arm P-D isolates that setting at one cell on 14 and 13 rows. A zero-shot characterization of a proposer therefore licenses no prediction about the same proposer inside a discovery loop without a measurement at that loop's own conditions.

What the arms found. Table 1 gives each arm that returned scoreable data, the bar it registered before sampling, and the outcome; Supplement S1 carries the full prediction codes. Two arms carry no row there. Arm GM finished UNDERPOWERED at 20 valid of 140, and arm GM2 returned 0 of 140 parseable at a 4,096-token output budget, the failure arm GM3 then diagnosed at 16,384.

Table 1: Six of the eight registered arms: what each changed about the setting, the bar registered before sampling, the measured outcome and the verdict. Prediction and falsifier codes are expanded in each arm's own section and listed in full in Supplement S1.

Arm	What changed	Registered bar	Outcome	Verdict
MU	The parent the proposer is conditioned on	Under a better in-family parent, anchor-rate at most 20% and keep-or-improve at least 50%	0 of 26 valid outputs keep the anchor; 26 of 26 keep or improve	F-MU1 triggered; the template dissolves
L	Five archive-and-mutate generations	Pooled conditioned on-prediction rate below the generation-0 rate in both cells (L2); falsifier F-L1	$N = 13$: 4 of 9 at generation 0, 5 of 20 conditioned. $N = 31$: 7 of 8, then 6 of 29	L2 holds; L1, L3 not met; L4 open (0 of 49 clear family)
P	The serving path: the bare pinned chat-completions endpoint	A floor of five valid outputs per cell	0 of 105 valid at the square cells	Every direct-emission prediction UNSCOREABLE
P-D	Extended thinking on against off, same pinned path	P-PD1 (the reasoning budget is the cause) against P-PD2 (the context is) against P-PD3 (anything else)	Thinking on, 12 of 14 valid with the template the plurality value at 5 of 12; off, 0 of 13	P-PD1 holds; extended thinking is the component
GM3	A second vendor's gemma at a 16,384-token budget	Pooled rate at least 30%, and the predicted value modal in 5 of 5 scoreable cells	46 of 80 pooled (57.5%); modal in 2 of 5; 11 of 43 at the discriminating cells	The pooled bar is met and the discriminating cells miss it
V	Thirteen free-tier aliases	At least 10 of a model's 25 invocations valid, then V1 at least 50% and V2 at most 10%	Ten aliases below the floor; 8 of 12 and 11 of 18 transfer at point estimate, 0 of 15 does not	The boundary is located, not attributed

Contributions.

1. *A measurement of transfer, not an argument about it.* Each change of setting was given its own registered arm, with numeric thresholds and a falsifier fixed before the first invocation. Eight arms are reported, and the deviations from every registration are tabled.
2. *A named instrument component.* The pinned-path collapse is not a system-prompt effect. One registered condition (P-D) and one post hoc diagnostic (temperature) separate the candidates, and extended thinking is the one that carries the template.

3. *A cross-vendor boundary that is located but not attributed.* Two second-vendor measurements disagree, with model variant and instrument confounded, so the boundary is located but not attributed.

Scope. The proposer under test is one vendor's weak-tier alias (`claude-haiku-4.5`), except where another vendor is named. Three trap cells carry the conditioning arm, $N = 13$, 21 and 31, and two of them, $N = 13$ and 31, carry the loop arm. At a trap cell the nearest-square grid must be truncated, so the template anchor sits strictly below the family's best value.

What is not claimed. Nothing about the optimizer, about what clears the recipe family, or about the Sonnet tier; those are the companion paper's results and are cited to it, not restated here (Anonymous authors, 2026). Nothing about mechanism inside the weights. Nothing about what any named discovery system does at generation zero, or about whether its diversity machinery (Mouret & Clune, 2015; Lehman & Stanley, 2011) repairs or explores. Preregistration is an adopted standard (Thomas et al., 2026; Vaccaro, 2026), not a contribution.

2 The template and its closed form

This section restates the object under test. Every definition in it is the companion paper's, cited to it (Anonymous authors, 2026), and nothing here is a result of this paper.

The benchmark. Maximize the sum of radii of N non-overlapping circles in the unit square. Feasibility is decidable exactly from emitted coordinates, so scoring needs no human. The direct-emission prompt fixes the count, states containment and non-overlap, forbids writing or executing code, and demands a raw Python list of `[x, y, r]` triples. The program channels replace the code-free clause with a program contract. The prompt is reproduced in the supplement with its digest.

The recipe family. Three terms are used throughout. The **recipe family** is the parametric set of grid-plus-filler constructions. A **template** is one member. The **anchor** is the template the distribution concentrates on at a given N.

A $k \times k$ grid places one circle per cell center with $r_{\text{grid}} = 1/(2k)$. Then m fillers sit on interior grid vertices, tangent to the four grid circles around them, with $r_{\text{filler}} = (\sqrt{2} - 1)/(2k)$ and $0 \le m \le (k - 1)^2$. Hence $V(k, m) = k/2 + m(\sqrt{2} - 1)/(2k)$. When $N < k^2$ the model does not drop to a smaller grid and fill it. It *truncates*, laying out the $k \times k$ lattice, occupying N cells and keeping the radius, which gives $T(k, N) = N/(2k)$.

The **family argmax** at N is the largest value the family can express there, recomputed in closed form over every admissible (k, m). Inside a trap zone it is the larger of $V(k^*{-}1, N - (k^*{-}1)^2)$, the drop-and-fill construction, and the truncated anchor $T(k^*, N) = N/(2k^*)$. At $N = 35$ the truncated grid is the larger of the two, which is why 35 is not a discriminating cell. The companion paper reports an independent linear program, blind to the recipe, that recomputes every closed-form value and aborts on disagreement. That verifier was not re-run over this paper's ledgers, which recompute the closed form directly, so no independent verifier covers the counts reported here.

The order rule and the branch rule. Write $k^*(N) = \text{round}(\sqrt{N})$ for the nearest-square order. This is a definition, not a hypothesis. The falsifiable content is the **branch rule**: $k^{*2} \le N$ gives extend with $m = N - k^{*2}$ fillers (converge), and $k^{*2} > N$ gives truncate the k^*-grid (trap). Substituting k^* into the branch condition gives the *trap zones* $N \in [k^2 - k + 1, k^2 - 1]$, which over the sweep range are $[13, 15]$, $[21, 24]$, $[31, 35]$, $[43, 48]$ and $[57, 63]$. At a converge cell the extend prediction is the argmax, so every discriminating cell is a trap cell, and the surviving in-family prediction is the truncate branch $T(k^*, N) = N/(2k^*)$.

Notation and conventions. The rows below are the companion paper's.

Term	Meaning
on-prediction	emitted sum within 2×10^{-3} of the registered predicted value
rival (argmax)	the highest-valued member of the recipe family at that N when it differs from the prediction: inside a trap zone, the drop-and-fill $V(k^*-1, m)$ wherever it beats the truncated anchor
discriminating cell	N where prediction $\neq$ family argmax; only these can distinguish anchoring from family search
validity	non-overlap and containment at 10^{-6} (primary) and 10^{-9} (logged), both always reported
trap cell	an N inside a trap zone, where the branch rule truncates; the discriminating cells are the trap cells where the truncated anchor sits below the family argmax
tier	nominal capability class of the serving alias addressed: here the weak tier is the vendor's Haiku alias (`claude-haiku-4.5`)
instrument	the proposer's serving path with the request parameters it carries, the reasoning setting included. Two are used here. The agent runtime does not expose decoding parameters and locks the conditioning context only in part. The pinned API is the vendor's chat-completions endpoint through OpenRouter, at temperature 1.0 with no system prompt. They are never pooled, and the word is used in this sense only
UNSCOREABLE	cell with fewer valid samples than its arm's registered floor; claims nothing
evaluability floor	the minimum valid-sample count a cell needs before its verdict counts, fixed per arm at that arm's registration and *not* common across arms: 3 for the second-vendor chain (arms GM, GM2 and GM3), 5 for arms P and P-D, and 10 valid of a model's 25 invocations for arm V
template anchor $T(k^*, N)$	the truncated-grid value at the nearest-square order, $N/(2k^*)$, the anchor of the truncate branch; at a discriminating cell it sits below the family argmax

3 Registration, instruments, scoring, floors

Registration protocol. For each cell the prompt is pinned and its SHA-256 digest written to disk first. The $N = 13$ square prompt digest opens `32db485b` and is given in full in the supplement. Predictions live in the header of each arm's analysis script or in a standalone registration file. Competing predictions, numeric thresholds, tie conventions and a falsifier are fixed before the first invocation. Raw outputs are stored verbatim, failures included. Scoring is deterministic and local. Each arm writes its rows to a JSONL ledger. The supplement carries the arm ledger of every arm with its design and outcome, the deviation table of every departure from a registration, and the per-cell tables for the serving-path and second-vendor arms.

Every prediction whose verdict this paper reports was registered before its arm's first scored run. Readings marked post hoc were not, and each is labelled where it appears. The protection against the garden of forking paths is registration and exhaustive reporting, not a family-wise error rate.

Two instruments, never pooled. The first is an agent runtime. It does not expose decoding parameters, so every runtime row is a distributional claim over an observed sampling regime. The subagent also inherits instruction files outside the task prompt, which the digests do not lock and a replicator cannot reconstruct. The second is a pinned API, the vendor's chat-completions endpoint reached through OpenRouter, at temperature 1.0 with no system prompt. It exposes and pins temperature and carries no inherited context. Arms MU and L run on the runtime. Arms P and P-D run on the pinned path. Arms GM, GM2 and GM3 run on a second vendor's first-party API, and arm V runs on free-tier aliases across seven attributed vendors. No count in this paper pools across instruments.

Two further properties are disclosed and not repaired. The alias-to-weights binding is a vendor promise rather than a hash, and hash coverage is incomplete where the supplement states.

Scoring conventions, fixed in advance. Validity is reported at 10^{-9} and 10^{-6}, both logged, with 10^{-6} primary. Proposers print six to eight decimals, so an eight-decimal tangency misses contact by about 5×10^{-9}, while 10^{-6} sits far below the gap of about 10^{-2} between rival constructions. Value matching uses a 2×10^{-3} window. That window never carries a clearance verdict. A proposal clears the family argmax when its verified sum strictly exceeds the closed-form argmax at the tolerance named with the count. Completions are parsed with `ast.literal_eval` after stripping code fences, and a completion failing to yield a list of N numeric triples is recorded as a parse failure and scored invalid, with the failure mode logged. Radii must be strictly positive. Every on-prediction rate is computed over valid outputs and so conditions on execution success. Per-cell validity denominators are reported beside every rate, except in arm MU, which registered pooled across its three cells and gives per-cell counts in its table where the registration scored per cell.

Floors are per arm and are not harmonized. The evaluability floor is the minimum valid-sample count a cell needs before its verdict counts. It is fixed at each arm's own registration. The second-vendor chain uses 3: a cell with fewer than three valid samples is UNSCOREABLE for mode analysis and claims nothing in either direction. Arms P and P-D use 5. Arm V uses a per-model floor rather than a per-cell one. A model needs at least 10 of its 25 registered invocations valid at 10^{-6}. Below that it is labelled BELOW-FLOOR and carries no anchoring claim in either direction. Arm L's registered floor is a two-part lineage rule: at least 3 valid proposals at generation 0, and at least 1 valid proposal in at least 3 of the 5 conditioned generations. All four lineages clear it. The 13-GREEDY lineage clears on 5 valid at generation 0 and on valid outputs in 3 of its 5 conditioned generations, so its 4 valid conditioned proposals are not the quantity the floor tests. Arm MU reports pooled rates with their denominators.

The eight arms. All eight were run once. Arms P and P-D are reported in full in both submissions; the other six are reported here and appear in the companion only as a pointer to this paper. What is unique here is the transfer framing and the clearance re-reads of arms MU and L. Arm MU issues one-parent mutation calls at three parents and three cells. Arm L runs a six-round archive-and-mutate loop at two cells by two archive regimes. Arm P sends the registered prompts through the pinned path at fifteen cells. Arm P-D sends the $N = 13$ prompt on that path under two one-line manipulations. Arms GM, GM2 and GM3 run the unconditioned-call protocol at a second vendor under three instrument settings. Arm V addresses thirteen free-tier aliases across seven attributed vendors.

4 One conditioning step: arm MU

A discovery loop does not call its proposer zero-shot. It hands the proposer a parent. Arm MU is the smallest version of that change: one parent, one instruction, no iteration.

Design. Arm MU issues one-parent mutation calls. The call is the bare prompt plus an existing packing plus the instruction to modify it so that the sum of radii increases. Three parents are used per cell. *A_anchor* is the model's own unconditioned anchor $T(k^*, N)$. *B_rival* is the provably better in-family rival $V(k^*{-}1, m)$. *C_offfamily* is a diagonal-chain packing scoring far below either. The cells are $N = 13$, 21 and 31, at $n = 15$ per condition-cell, which is 135 invocations. The arm was registered before sampling with numeric decision thresholds, a power analysis, tie-against conventions, and a falsifier. The falsifier's consequence was written down before the first invocation: if it fires, the anchoring result is scoped to unconditioned calls only.

Registered predictions. P-MU1 predicted an anchor-rate under *A_anchor* of at least 50%. P-MU2 predicted an anchor-rate under *B_rival* of at least 35%. P-MU3 predicted k-inheritance under *B_rival* below 50%. P-MU4 predicted that under *C_offfamily* the modal valid output sits on the k^*-grid, scored per cell. The registered decision rule reads DISSOLVES when the pooled anchor-rate under *B_rival* is at most 20% and the keep-or-improve rate is at least 50%. Falsifier F-MU1 fires on the same condition.

F-MU1 triggered. Under *B_rival* the pooled anchor-rate is 0 of 26 valid outputs. That is below the registered dissolution threshold. Keep-or-improve is 26 of 26, above its bar. Both halves of the rule are therefore met. F-MU1 is the negation of the persistence prediction and fires on exactly that pair of conditions, so it is reported for completeness and not as a second result. Every valid *B_rival* sample keeps or refines the rival construction. Every one also inherits the rival's grid order, so k-inheritance is 26 of 26, and P-MU3 is disconfirmed. The template does not reassert itself when a better in-family parent is present.

P-MU1 and P-MU2 were not met. *A_anchor* makes the same point from the other side. The parent is the model's own anchor and the instruction asks for a larger sum. Only 7 of 28 valid samples stay on the anchor. P-MU1 predicted at least half. P-MU2 predicted at least 35% under *B_rival*, and the observed rate there is zero.

P-MU4 held in two cells of three. The one condition where the template does reassert itself is the one the registration predicted it would. Given the off-family diagonal parent, 20 of 34 valid samples abandon the parent entirely and emit a k^*-grid construction. Per cell the prediction holds at $N = 21$ and at $N = 31$. It does not hold at $N = 13$, where the count is 1 of 7. The attractor is real and weak: it captures the trajectory only when the parent is worse than what the template supplies.

Parent	Valid	Anchor-rate (CI)	Registered threshold	Keep-or-improve	k-inher.
B_rival (in-family argmax)	26	0/26 (0%, $[0.00, 0.13]$)	DISSOLVES if $\leq 20\%$	26/26 (bar $\geq$ 50%)	26/26
A_anchor (model's own anchor)	28	7/28 (25%, $[0.13, 0.43]$)	P-MU1 predicted $\geq 50\%$	—	—
C_offfamily (diagonal chain, sum ≈ 0.6)	34	20/34 abandon parent for k^*-grid (58.8%, $[0.42, 0.74]$); per cell 1/7, 7/13 and 12/14 at $N = 13$, 21 and 31	P-MU4: template reasserts, scored per cell	—	—

No conditioned output clears the family argmax at 10^{-9}; the eighteen that clear it at 10^{-6} are overlapping packings (post hoc, disclosed). Arm MU's rows were scored against the anchor, so the clearance question was never asked at registration. It is asked here, and the reading is post hoc. Recomputing the family argmax from the closed form gives 1.776142375 at $N = 13$, 2.258883476 at $N = 21$ and 2.748528137 at $N = 31$. Arm MU's 135 invocations give 88 valid outputs at the primary tolerance. They give 63 valid at 10^{-9}. Eighteen of the 88 sit above the family argmax by more than the 5×10^{-8} granularity at which their own sums are recorded, and by at most 1.5×10^{-6}. Every one of the eighteen fails the 10^{-9} test. They are packings whose overlap is small enough to pass the loose gate, and the excess is the overlap. Twelve further rows exceed the argmax only within that recording granularity, which is to say they are the argmax written down, and the remaining 58 of the 88 do not exceed it. At 10^{-9} none of the 63 clears.

5 Five loop generations: arm L

Arm MU samples once. The regime a discovery loop actually runs in iterates. A proposal is scored, kept or discarded, and fed back as a parent. Arm L runs the minimal generational loop and reports its four registered predictions whether or not they held.

Design. The design was registered before sampling, with both prompt templates hashed and the stepper and scorer committed alongside. Generation 0 is the bare template, byte-identical to the unconditioned arms of the companion paper. Generations 1 to 5 use arm MU's registered mutation template verbatim, instantiated with the parent packing and its score. Two discriminating cells, $N = 13$ and $N = 31$, are crossed with two archive regimes. GREEDY retains only the single best valid proposal. DIVERSE retains up to five distinct-valued ones. That gives four lineages of six rounds at population 5, which is 120 invocations. Two were runtime-rejected and counted. The parent for population slot i is archive[i mod |archive|], so the loop carries no sampling randomness of its own. All four lineages are evaluable under the registered floor.

This is a minimal analogue and the registration says so. Population 5, six rounds, one-parent mutation, no islands, no crossover, no program representation, and no evaluator signal beyond the parent's own score. Nothing here transfers to deployed systems, and no claim about any named system follows.

Registered predictions. L1 predicted that the modal valid output at generation 0 equals the registered prediction in every lineage. L2 predicted that dissolution survives iteration, so that the pooled on-prediction rate over generations 1 to 5 falls below the generation-0 rate. L3 predicted that the pooled conditioned on-prediction rate is higher under GREEDY than under DIVERSE, on the reading that loop diversity is scaffold-produced. L4 was registered open with no direction predicted: report best-of-run against the family argmax. Falsifier F-L1 fires if the pooled conditioned rate meets or exceeds the generation-0 rate in both cells.

L2 confirmed. Dissolution survives iteration in both cells. Summing the two lineages at each cell in Table 2, the on-prediction rate falls from generation 0 to the conditioned rounds at both N. F-L1 did not fire.

Table 2: Arm L by lineage. Valid and on-prediction counts at generation 0 and across the five conditioned generations, with each lineage's best scoring valid output and its cell's family argmax. The pooled row sums the four lineages.

Lineage	Gen-0 valid	Gen-0 on-prediction	Conditioned valid (gens 1 to 5)	Conditioned on-prediction	Best of run	Family argmax
13-GREEDY	5	2	4	0	1.7761424	1.7761424
13-DIVERSE	4	2	16	5	1.7761424	1.7761424
31-GREEDY	4	3	18	3	2.6104167	2.7485281
31-DIVERSE	4	4	11	3	2.7485281	2.7485281
Pooled	17	11	49	11		

L1 was not met. It fails on one lineage. The 13-DIVERSE seed round in Table 2 splits evenly, half its valid proposals on the prediction and half elsewhere, and the registered tie rule counts the tie against the prediction. The other three lineages pass. The registered rule set four separate per-lineage bars over about four valid proposals each, rather than one pooled bar per cell, and the miss is a miss of that rule. It is reported as scored and not rescued.

L3 was not met, with the direction reversed. The conditioned on-prediction rate is 3 of 22 under GREEDY. It is 8 of 27 under DIVERSE. The comparison is confounded. Greedy and diverse lineages differ in parent identity as well as archive width, and 13-GREEDY contributes the fewest conditioned valid proposals of any lineage in Table 2. The reversal is reported as a disconfirmation of the registered prediction and not as evidence for its converse.

L4: three of the four lineages reach the family argmax exactly, and none exceeds it. The best-of-run and family-argmax columns of Table 2 carry the comparison: both lineages at $N = 13$ and 31-DIVERSE land exactly on their cell's family argmax, and 31-GREEDY stops short of it. Across the valid conditioned outputs the table counts, zero exceed the best value the recipe family can express at 10^{-9}, the tolerance the clearance test uses. Iteration moves the proposer off the anchor without moving it past the family in this arm. One correction is disclosed. The registered configuration stored the family argmax to seven decimals. The clearance test compares at 10^{-9}. A proposal landing exactly on the rival at $N = 31$ therefore tested as 3.7×10^{-8} above the truncated literal. The scorer now recomputes the argmax in closed form; that proposal agrees with the recomputed value to 1.3×10^{-15}, float noise, and clears nothing. The change is post-sampling and affects only the 10^{-9} clearance test.

Scale is the honest limit. Five conditioned generations, population 5, four lineages whose valid counts Table 2 reports, one tier. The arm shows that this loop, at this size, on this channel, does not restore the template and does not clear the family. It cannot show that a larger one would not.

6 Serving path and reasoning budget: arms P and P-D

Arms MU and L change what the proposer is conditioned on. Arms P and P-D change how it is served. Both were registered before sampling. Their per-cell tables are in Supplement S5. Both arms were run once, and this paper and the companion report the same rows (Anonymous authors, 2026).

Arm P: the registered prompt on a bare, pinned serving path. Arm P sends the registered prompts byte-identical through a serving path that exposes and pins temperature. It runs at $n = 15$ per cell across seven square cells, five held-out cells and three code cells, which is 105, 75 and 45 invocations and 225 in all. Of those, 219 returned content and 6 were refused by the serving path and counted invalid at the cell they were issued for. None of the six was a square cell, and all six are listed by cell in Supplement S5. The arm was registered with competing predictions on mode identity (P-P1, P-P2), on the code-channel clearance (P-P3, P-P4), and a falsifier on the discriminating-cell rate (F-P1). Validity collapsed. At the seven square cells, 0 of 105 outputs are valid at either tolerance, and all 105 returned content. Every failure but three is an overlap, the three being one non-positive radius, one count mismatch and one malformed row, at $N = 35$, 37 and 43 in that order. The responses are well-formed lists of the requested count, with no prose and no fences, emitted with no reasoning tokens. Their median maximum pairwise overlap at $N = 13$ is 0.10, which is fifty times the value window. At the held-out cells 4 of 75 are valid, all four at $N = 50$ and all four summing to 2.5, a full unit below the family argmax there. None clears. The direct-emission predictions are therefore UNSCOREABLE at every cell under the arm's own floor of five valid outputs, and F-P1's denominator is empty. The registration did not anticipate a collapse, and that is reported rather than read as a zero either way. On the code cells there are 12 valid programs of 45, by cell 4, 6 and 2, with only $N = 21$ above the floor. None clears the family, so P-P3 holds. The per-cell rows are in Supplement S5. Two post hoc diagnostics, disclosed as such, bound the cause. At temperature 0.0, 0.5 and 1.0 the $N = 13$ prompt gives 0 of 6 valid at each, which is 0 of 18 across the three settings. The same prompt through the agent runtime on the same day gives 6 of 6 valid and lands the registered anchor $T(4, 13) = 1.625$ in 4 of 6.

Arm P-D: which component of the instrument. Arm P-D was registered after an outside review asked for a diagnostic beyond temperature, and the registration was pushed before the first call. It sends the same $N = 13$ prompt, to the same alias, on the same pinned path and decoding, under two one-line manipulations, at $n = 15$ each. D1 turns the vendor's extended thinking on at the routing layer's default effort, with no system prompt. D2 leaves thinking off and adds one system line asking for care. The predictions compete. P-PD1 says the reasoning budget is the cause, and predicts D1 valid in at least 8 of 15 with D2 valid in at most 3. P-PD2 is the mirror image, on which the context is the cause. P-PD3 covers anything else. The arm closed at 27 of 30 rows by a registered amendment, after the serving path refused the last three for credit. Those rows could not have changed the verdict, because D2 could reach at most 2 valid while D1 already held 12. D1 is valid in 12 of 14. D2 is valid in 0 of 13, all overlaps, which is exactly arm P's collapse. P-PD1 holds. Validity returns, and the template is the plurality value: $T(4, 13) = 1.625$ at 5 of 12 valid D1 outputs, and grid order $k = 4$ is modal at 6 of 12, so the secondary prediction S-PD1 is met. The named component is extended thinking, on against off at the routing layer's default effort, a request parameter on the pinned path. Only $N = 13$ was run under D1 and D2, at 14 and 13 rows, so the component is named on one cell and no interval is computed on those counts.

7 Other vendors: arms GM, GM2, GM3 and V

A fourth change of setting is the vendor. Four registered arms make it. Three of them, GM, GM2 and GM3, run against a second vendor's first-party API with pinned prompts, a fixed output budget and a logged provider per row, and vary the model family. The fourth, arm V, varies vendor and serving class at once. The per-cell tables and the chain's verdict table are in the supplement.

Arms GM and GM2 returned no scoreable data, and both are reported. The chain registration fixes one set of definitions and predictions for all three arms. A cell with fewer than three valid samples is UNSCOREABLE. The primary prediction requires a mode match in at least 5 scoreable cells. The second requires a pooled on-prediction rate of at least 30%. The third requires the two-radius signature in at least half of on-prediction samples: at order k^* that is a grid radius $1/(2k^*)$ carrying a filler radius $(\sqrt{2} - 1)/(2k^*)$, and the rival's form is the same pair at order $k^* - 1$. The falsifier fires on a mode-match failure in at least 4 scoreable cells. With 5 scoreable cells the pass bar needs 5 of 5 and the falsifier 4 or more failures, so 1 to 3 failures are unadjudicated by the registration. Arm GM addressed a gemini-flash-lite alias at seven cells by twenty samples. It returned 20 valid of 140, fewer than three per cell on average against the chain's floor of three, and was declared UNDERPOWERED against those bars. Arm GM2 addressed the gemma family at a 4,096-token output budget. It returned 0 of 140 parseable. The cause is a budget confound rather than a model result: the vendor API returns deliberation on a separate channel, and the smaller budget died inside it. Arm GM3 re-ran the same design at 16,384 tokens, so GM2 and GM3 differ only by output budget at one vendor, and the same model parses at 123 of 140. Of the 140 sampled rows, 17 did not parse, 123 did, and 80 of those are valid packings at 10^{-6}, leaving 43 that parsed and then failed validity. A cross-vendor null under a fixed output budget is not evidence about the model until the budget's interaction with the vendor's response structure is checked.

Arm GM3: the pooled bar is met and the discriminating cells are not. Arm GM3 runs the unconditioned-call protocol against gemma-4-26b through the vendor's first-party API, at seven N cells by twenty samples. Predictions and falsifier were fixed before sampling. One serving-path deviation is registered

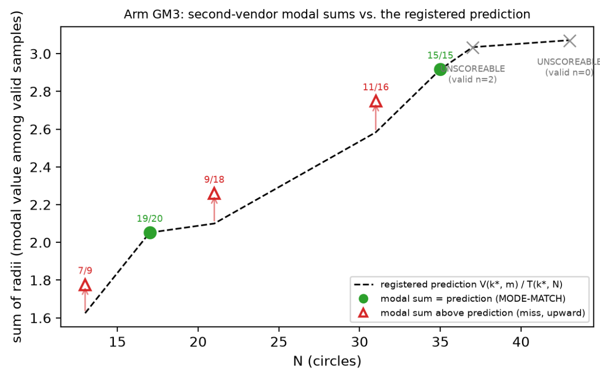

Figure 1: *Arm GM3 modal sums against the registered prediction.* Filled green circles match the predicted mode (the dashed line is $V(k^*, m)$ at extend cells and $T(k^*, N)$ at truncate cells), with frequencies annotated. Open red triangles miss, and every miss lands above the prediction at the rival argmax *value*. That these are the rival *construction* is established by the post hoc structural check in the text, not by this figure. The two UNSCOREABLE cells are drawn as grey crosses on the prediction line, annotated with their valid counts, so the panel carries seven marks and not five. Regenerates via `fig_gm3_anchoring.py` from `arm_gm3_report.json`.

and disclosed. The first-party key stalled on quota, so 41 of the 140 rows were sampled through a routed path, with the provider logged per row. The amendment was committed before the resumed sampling. Both accountings are reported. The pooled prediction held. Of the 80 valid packings, 46 land within 2×10^{-3} of the value the closed form predicts, against a registered 30% bar. That rule was fitted entirely on another vendor's models before this family was ever sampled. The 41 routed rows carry 2 of the 80 valid packings and none of the 46 on-prediction, so first-party rows alone give 46 of 78 (59.0%) against the mixed set's 46 of 80 (57.5%), and no verdict changes. The pass rides on the non-discriminating cells. 35 of the 46 on-prediction packings sit in the two cells where prediction equals the family argmax. On the three scoreable discriminating cells the rate is 11 of 43, below the bar the pooled metric passes. The cell-level prediction did not hold: the predicted value is modal in 2 of 5 scoreable cells, short of the registered 5, and Figure 1 plots the per-cell modal sums against that prediction. The falsifier did not trigger either, because it needed failure in at least 4. The pass bar and the falsifier leave outcomes of one to three failures unadjudicated, and the observed three failures fall there. The three misses are misses of the registered rule. We report the verdict that rule defines and note that a registration with a gap this wide buys less than its two bars imply. The structural prediction came in under its bar: 20 of 46 on-prediction samples carry the exact two-radius decomposition, against a bar of half. The signature test is blind by construction at truncate cells, whose prediction is single-radius.

The misses are the rival construction (post hoc, disclosed). The three cell-level misses are not noise. The family's modal output at every discriminating cell is the rival argmax itself. A post hoc structural diagnostic, read-only over the ledger, finds all 27 of 27 rival-valued packings carrying the rival's exact two-radius decomposition, which is the $(k^*{-}1)$ grid radius with the $(k^*{-}1)$ filler radius. The registered signature test could not see this, because it tests radii against the predicted order k^*. On the discriminating cells 27 of 43 valid packings are the rival construction. Validity collapsing at the two largest cells bounds the family's competence window on this task, and both cells are UNSCOREABLE under the registered rule and claim nothing.

Arm V: thirteen free-tier aliases. Arm V varies vendor and serving class together, under the same byte-identical prompts arm V and the GM chain both inherit, at five cells, $N = 13$, 17, 31, 35 and 37, and five samples per model and cell, against free-tier aliases across seven attributed vendors. The registration was committed before sampling. Three amendments, each committed before the sampling it governs, expanded the original three-alias design to thirteen aliases across two serving paths, and raised the output budget after hidden-reasoning truncation emptied three models' replies. The registered floor is at least 10 of a model's 25 registered invocations valid at 10^{-6}, and below it the arm makes no anchoring claim in either direction.

Execution retried each invocation slot until content arrived, per the registered transport-repair rule, so the ledger holds many more attempt rows than slots. There are 877 attempt rows over 325 registered slots, of which 314 carry a row, and the floor is evaluated on the slots, so a slot with no row counts against the floor. Ten of thirteen aliases finished BELOW-FLOOR, including all three originally registered aliases, and one of them returned all 108 of its attempts without content. Three aliases cleared the floor, and their registered verdicts split. V1 predicted at least 50% of a model's valid outputs on-prediction, and V2 at most 10% at the rival argmax. V2 is scored only at the discriminating cells, $N = 13$ and 31. All three come from the amendment-added set, so every transfer verdict in this arm rests on aliases registered by amendment.

alias	vendor	valid	V1 on-prediction	V1 verdict (bar: ≥50%)	V2 rival-argmax valid at discriminating cells (bar: ≤10%)
`north-mini-code`	Cohere	12	8/12 [39–86%]	**TRANSFERS** (point est.)	0/4 **HOLDS**
`gpt-oss-20b`	OpenAI	18	11/18 [39–80%]	**TRANSFERS** (point est.)	0/9 **HOLDS**
`gemma-4-31b-it`	Google	15	0/15 = 0%	**DOES-NOT-TRANSFER**	0/3 **HOLDS**

V2 is scored only at the discriminating cells. Verdict labels are the registration's; both passing intervals span the 50% bar.

Both passes are weaker than the percentages suggest, in three stated ways. The bracketed Wilson 95% intervals include the bar, so each verdict is a point-estimate pass at $n = 12$ and $n = 18$. Three of thirteen aliases were evaluable at all. And the pooled rate spans cells that cannot discriminate anchoring from family search. Decomposing onto the discriminating cells gives 3 of 4 for the first alias and 5 of 9 for the second. Those denominators are valid outputs at those cells and not registered slots. Applying the template-shape null costs the arm its strongest reading. That null is a proposer choosing uniformly among one branch value per grid order $k \in \{k^*-1, k^*, k^*+1\}$, which lands on-prediction one time in three, and Supplement S5 restates it with the tails (Anonymous authors, 2026). The pooled rates clear that null, at exact upper tails of 0.019 and 0.014. They clear it on cells where prediction equals the family argmax. At the cells that can discriminate, neither pass separates: 3/4 gives 0.111, 5/9 gives 0.145, and arm GM3's 11/43 gives 0.895. The honest cross-vendor statement is that two further vendors are consistent with the behaviour, and that at these sample sizes the discriminating evidence does not separate them from a shape-uniform proposer. The strong form holds everywhere evaluable within this arm: zero rival-argmax emissions among the 16 valid outputs at discriminating cells.

The boundary runs both ways at the pooled level. At the discriminating cells neither direction is established at these sample sizes. Two further vendors' models land the registered construction values at point estimates above the bar, so anchoring under this instrument is not confined to the original family at the pooled level, and the discriminating cells do not settle it. It is also not universal. Under the chain's first-party instrument, gemma-4-26b emits the rival construction in 63% of its valid outputs at discriminating cells. Under arm V's free-tier instrument, gemma-4-31b lands neither the prediction nor the rival. Its valid packings are weaker de-novo constructions whose per-cell bests run 0.88–1.85 against per-cell predictions of 1.63–3.03, so the best valid packing falls below the prediction at every one of its cells. Model variant and instrument quality change together across those two measurements, so which difference carries the split is not attributable from the pair. Either way, no single-model measurement would have seen the boundary.

8 Related work

The loop the proposer is characterized for. Language Model Crossover (Meyerson et al., 2024), ELM (Lehman et al., 2022), EvoPrompt (Guo et al., 2024), EoH (Liu et al., 2024) and LLaMEA (van Stein & Bäck, 2025) established the LLM call as an evolutionary operator over prompts and programs. FunSearch (Romera-Paredes et al., 2024) turned the pattern into a discovery claim, and AlphaEvolve (Novikov et al., 2025) and its successors inherit it, OpenEvolve (Sharma, 2025) among them (Lange et al., 2025; Su et al., 2026; Khrulkov et al., 2025; Cemri et al., 2026; Wang et al., 2025). Every one of them conditions its proposer on a parent and iterates it, which are two of the conditions this paper varies. Simple baselines (Gideoni et al., 2026) and classical solvers (Berthold et al., 2026) recover much of the reported advantage.

Template convergence under iteration. "Mutation Without Variation" (Gurkan et al., 2026) finds iterated LLM program mutation collapsing onto previously seen structural templates: in 87% of chains, over 93% of mutations revisit a seen form. That work runs mutation loops in program space with no closed form and no preregistration, which makes it the closest published neighbour to arm L. Their result is about structural revisiting, and arm L's second registered prediction is about a value the revisited structures do not cross.

Anchoring, and what kind of object anchors here. Anchoring is studied as a general bias over initial information and as numeric primes dragging point estimates (Huang et al., 2026; Lou & Sun, 2024; Malberg et al., 2025). The modality here differs, because a construction template has a computable value where a scalar anchor has only a measurable pull. One level up, LLM outputs concentrate where human ones do not (Chen et al., 2026; Jiang et al., 2025). Work on set-level and diversity-aware selection (Li, 2026; Wang et al., 2026; Yang et al., 2026) and the bin-packing critiques (Herrmann & Pallez, 2026; Sim et al., 2025) converge on the same question about the proposer's contribution.

Two results that cut against this paper's framing. Evolutionary search matters empirically (Zhang et al., 2024), so a single unconditioned call is a poor stand-in for a loop, and strong LLM optimizers act as local refiners (Zhang et al., 2026b), which is what a parent-conditioned call is.

Instrument effects and reasoning budget. Larger and more instructable models become less reliable (Zhou et al., 2024), a reasoning model can report a higher hallucination rate alongside higher accuracy (OpenAI, 2025), and reasoning collapses past a complexity threshold (Shojaee et al., 2025). Geometric construction is a regime where nominal capability and executed correctness diverge (Kim et al., 2026; Balunović et al., 2025).

Preregistration lineage. Recipes, outcome-blinded predictions and agent protocols are already preregistered (Thomas et al., 2026; Williams et al., 2026; Vaccaro, 2026; Zhang, 2026), and HindsightBench (Jia, 2026) releases frozen preregistrations of directional aggregate hypotheses. This paper's arms lock a persistence prediction with a numeric threshold and a falsifier, registered before the setting was changed.

9 Limitations, reproducibility, deviations

Scale. Arm MU takes one conditioning step and arm L runs five generations of a registered archive-and-mutate loop, so scale stays open. Two small arms meet the evidence that evolutionary search contributes (Zhang et al., 2024) and that strong LLM optimizers act as local refiners (Zhang et al., 2026b) without settling it.

Vendor and tier scope. The proposer under test is one vendor's weak-tier alias except where another vendor is named. The cross-vendor arms move the pooled-level statement across vendors and not the cell-level mode structure, which is confirmed only within the original family. Ten of thirteen aliases in arm V finished below their floor and claim nothing.

Instrument. The agent runtime does not expose decoding parameters, so arms MU and L are distributional claims over an observed sampling regime. The alias-to-weights binding is a vendor promise, so no result here is a statement about a released model version. The prompt digests lock the task prompt and not the inherited conditioning context, which is the hole arms P and P-D were registered to probe.

Prompt robustness. Every arm conditions on a single hash-locked prompt, which buys reproducibility and not robustness to rewording. The companion paper reports a paraphrase arm at two discriminating cells (Anonymous authors, 2026).

Untested alternatives. Typicality bias in preference-tuned models (Zhang et al., 2026a) predicts concentration on the most typical construction without reference to geometry, and no observation here excludes it. Contamination probes (Carlini et al., 2021; Golchin & Surdeanu, 2024), a format-sensitivity sweep (Sclar et al., 2024) and alignment-induced diversity loss (Kirk et al., 2024) as the account of the cross-vendor concentration differences all remain untested.

Reproducibility. Every arm's prompts were hashed and its predictions and falsifiers fixed before its sampling, with the exceptions the deviation table in the supplement records. No arm was selected on its result, and every arm is reported. The complete artifact accompanies this submission as anonymized supplementary material: ledgers, scripts, registrations, amendments and frozen reports, from which every number regenerates by a documented command. Arms P and P-D can be re-sampled, because they run on a path whose parameters are pinned. Arms MU and L can be re-scored and cannot be re-drawn, because their rows depend on a dispatch wrapper and inherited context the digests do not lock. The post hoc readings carry no multiplicity correction and are descriptive.

Deviations, in summary. Full rows are in the supplement. Arm GM3 has one serving-path deviation, registered before the resumed sampling, covering 41 of 140 rows, with both accountings reported. Arm V has three amendments, each committed before the sampling it governs, which expanded the alias set, repaired transport, and raised the output budget. Arm L discloses one post-sampling scorer correction that affects only the 10^{-9} clearance test. Arm P-D closed at 27 of 30 rows by a registered amendment, and the remaining rows could not have changed its verdict.

Use of AI systems. Claude models drafted manuscript prose and assisted implementation. The human author designed the study, specified the instruments and their decision rules, and approved each preregistration before sampling. Each preregistration commit is a git ancestor of the sampling it governs, which is what a reader can check.

10 Conclusion

Eight registered arms changed four things about the setting a weak-tier proposer is called in: the parent, the iteration, the serving path with its reasoning setting, and the vendor. One conditioning step dissolved the template on both halves of the registered rule. Five loop generations did not restore it at this scale. A change of serving path collapsed validity until extended thinking was turned on at the routing layer's default effort. At a second vendor the pooled prediction held and the discriminating cells did not. The template a weak-tier LLM proposer concentrates on in zero-shot circle packing is contingent on the serving path and the request parameters it carries, the reasoning setting included; it is neither confined to one instrument nor invariant across them, and arm P-D isolates that setting at one cell on 14 and 13 rows. A zero-shot characterization of a proposer therefore licenses no prediction about the same proposer inside a discovery loop without a measurement at that loop's own conditions. A study that reasons about such a characterization inside a loop owes that step a measurement, and here each step cost one arm.

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